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Session : 26208

Integrated Ring-Based Network Architecture for IT and OT Service Segregated

Presented By

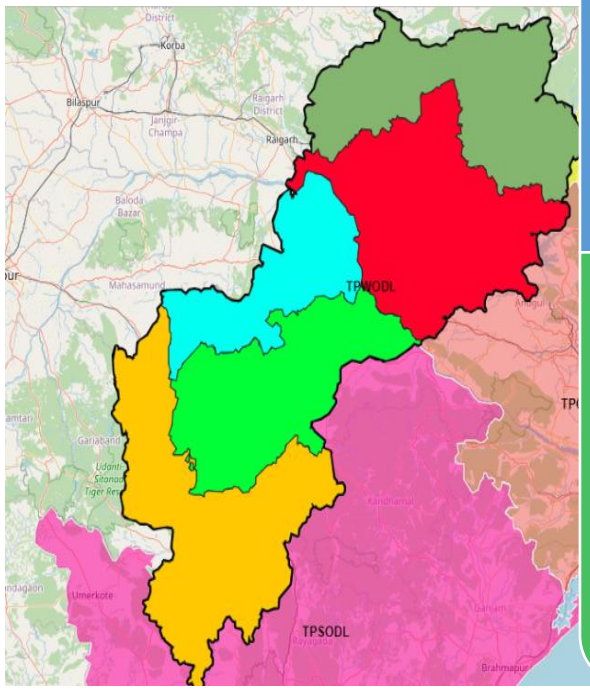
Pritesh Kumar Srivastava, HoG Communication

TP Western Odisha Distribution Limited

PRESENTATION GUIDELINES

- **Presentation Time: The total time for presentation is 12 mins. You are requested to be present on-time as per your session slot.**
- Presentation Format: All the presenting speakers are required to give a presentation as per the PPT Format. (Maximum 7 Slides). Please note that the additional presentation slides cannot be added in the shared PPT format.

INTRODUCTION



TPWODL

88 lakh Population Served

48,373 Sq. Kms of Distribution Area

21.42 lakh Customer Base

Reliability improvement is one of the key KPI

Responsible for end-to-end network connectivity for both IT Offices & PSS locations through MPLS /4G/OFC

Establish reliable, secure and stable network connectivity to APSCC

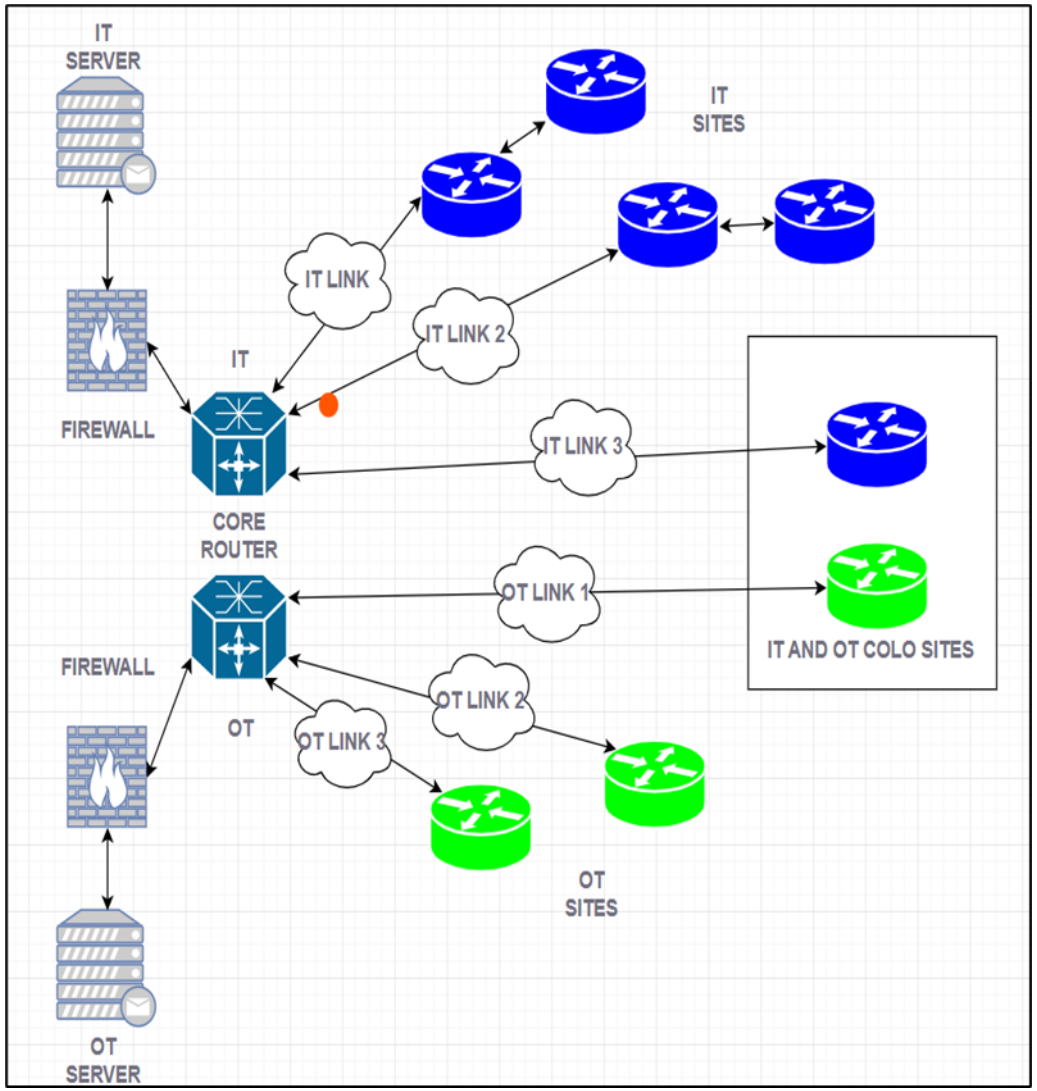
Provide network connectivity to Smart Metering solution through IoT & MPLS

Provides Integrated Communication Network (ICN) connectivity with other three DISCOMs through OPGW

Communication department is backbone for the SCADA & IT Connectivity across all the offices

The entire success of remote operations of PSS depends on the QoS of communication network

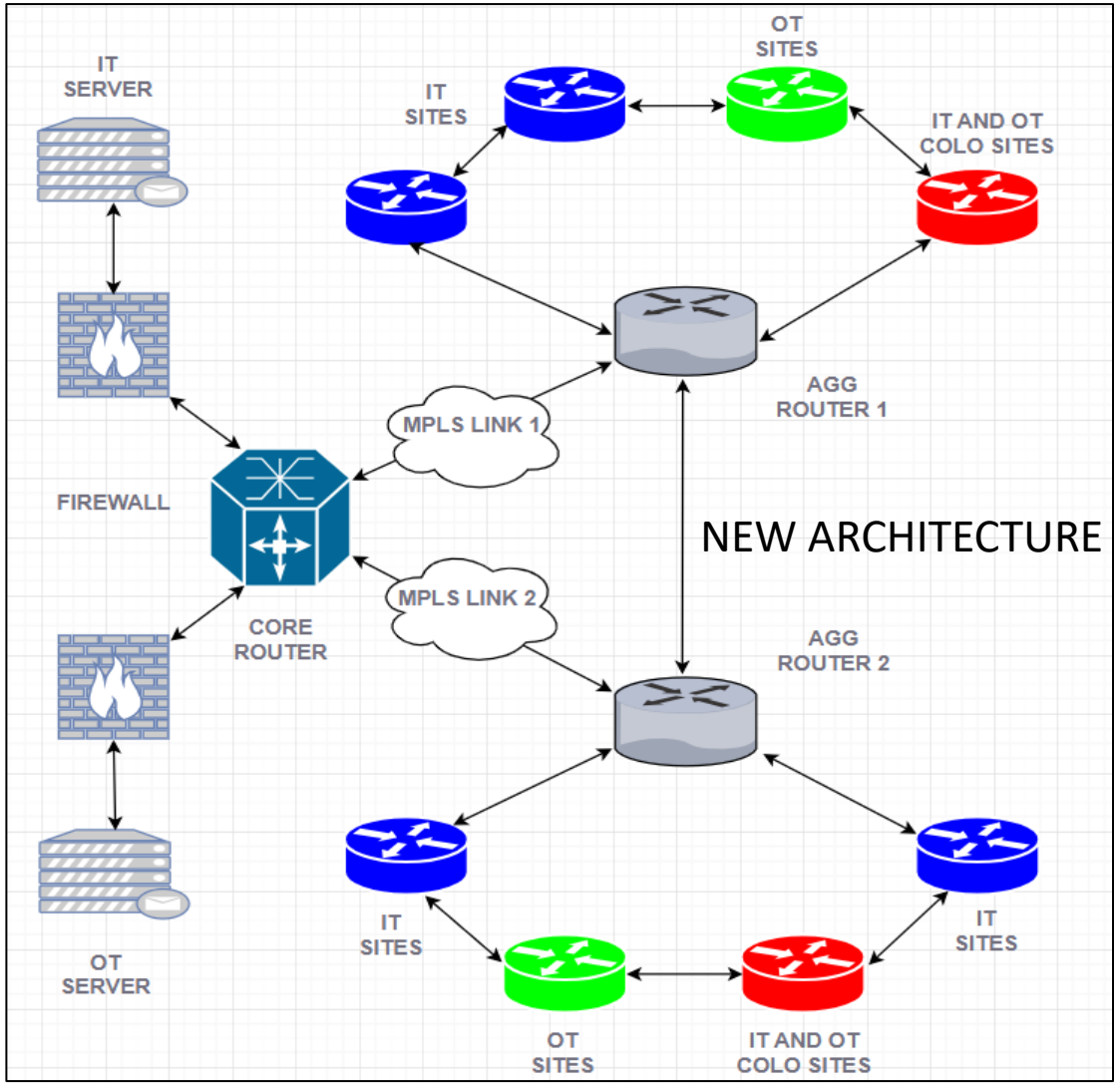
CONTEXT



PRE ARCHITECTURE

Separate Physical Networks	No Redundancy	Demerit
• IT and OT services ran on independent physical hardware. • This resulted in higher CAPEX, duplicate O&M, and complex vendor management.	• Most sites depended on single MPLS links or fiber. • Any fiber cut, MPLS outage, or router failure led to complete site isolation. • OT services like SCADA and unavailability critical alarms resulted to high risk during failures.	• High Cost for Multiple MPLS Link • Reduced Network Uptime • Customer dissatisfaction

RELEVANCE



NEW-ARCHITECTURE

- Redundant Architecture
- All access sites are connected in ring architecture with Aggregation router.
- Each ring consists of both IT and OT sites with same physical path but virtually segregated through dual VRF concept.
- Use of single router for IT and OT in co-located sites.
- MPLS link delivered at Aggregation nodes only.

STEPS FOR IMPROVEMENTS

VRF SEGREGATION

- IT VRF used for enterprise apps, internet egress, cloud access
- OT VRF used for SCADA & PSS protection alarms.

RING TOPOLOGY

- Fiber ring interconnects office locations as access nodes with bidirectional traffic flow.

MPLS REDUNDANCY

- Each Aggregation node dual-homed via MPLS for redundancy.
- Automatic reroute on failure.

PRESENTATION ON THE TOPIC (2/2) (1 Min)

High Availability

- Physical ring topology & Dual MPLS paths ensure service continuity under link/node/site failures.

Security & Segmentation:

- Separate VRFs enforce strict IT/OT isolation.
- Use of GRE Tunnel for End to End encryption.

Cost:

- Single Router used for IT & OT services leads to cost optimisation.

USE CASE / CASE STUDY

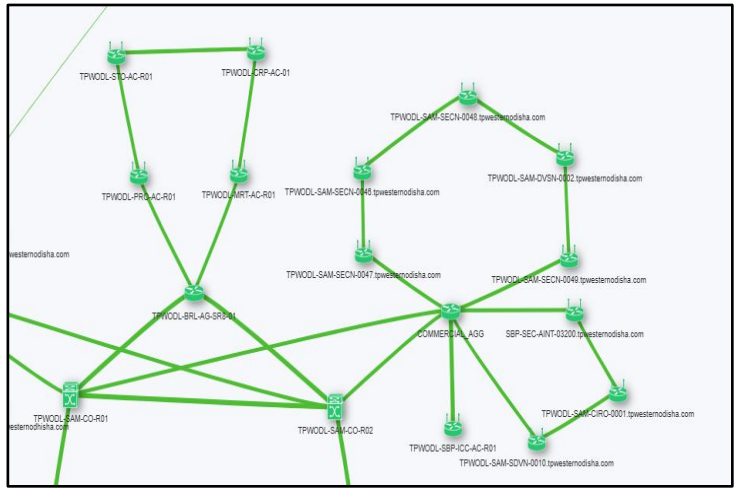
TPWODL-SAM-CIRO-0001.tpwesternodisha.com - Gi0/0/3

SL. NO.	DATE	HOSTNAME	IP ADDRESS	ASSET ID	NAME	NETWORK AVAILABILITY (AVERAGE)
1	Jan 2026	TPWODL-SAM-CIRO-0001.tpwesternodisha.com	10.20.8.1	ASSET0184	TPWODL-SAM-CIRO-0001.tpwesternodisha.com - Gi0/0/3	99.98 %
2	Feb 2026	TPWODL-SAM-CIRO-0001.tpwesternodisha.com	10.20.8.1	ASSET0184	TPWODL-SAM-CIRO-0001.tpwesternodisha.com - Gi0/0/3	99.97 %

TPWODL-SAM-SDVN-0010.tpwesternodisha.com - Gi0/0/0

SL. NO.	DATE	HOSTNAME	IP ADDRESS	ASSET ID	NAME	NETWORK AVAILABILITY (AVERAGE)
1	Jan 2026	TPWODL-SAM-SDVN-0010.tpwesternodisha.com	10.20.17.1	ASSET0183	TPWODL-SAM-SDVN-0010.tpwesternodisha.com - Gi0/0/0	100 %
2	Feb 2026	TPWODL-SAM-SDVN-0010.tpwesternodisha.com	10.20.17.1	ASSET0183	TPWODL-SAM-SDVN-0010.tpwesternodisha.com - Gi0/0/0	99.97 %

NETWORK UPTIME FROM NMS IN ACCESS SITES



NETWORK TOPOLOGY FROM NMS

```
A:COMMERCIAL_AGG# show router 10002 interface

===============================================================================
Interface Table (Service: 10002)
===============================================================================
Interface-Name                   Adm       Opr(v4/v6)  Mode    Port/SapId
   IP-Address                                                   PfxState
-------------------------------------------------------------------------------
Towards AITHAPALI_SECTION        Up        Up/Down     VPRN    1/2/1:0
   10.190.199.97/30                                             n/a
Towards SDO Aintapli             Up        Up/Down     VPRN    1/2/2:0
   10.190.199.110/30                                            n/a
Towards-Airtel                   Up        Up/Down     VPRN    1/3/1:0
   10.20.108.33/27                                              n/a
Towards_ESO_Khetrajpur           Up        Up/Down     VPRN    1/2/3:0
   10.190.199.26/30                                             n/a
Towards_ESO_PHD                  Up        Up/Down     VPRN    1/1/1:0
   10.190.199.5/30                                              n/a
loopback-1                       Up        Up/Up       VPRN    loopback
   10.190.205.17/32                                             n/a
   fdd5::8000:0:0:d11/128                                       PREFERRED
   fe80::4f:75d4:21a2:af53/64                                   PREFERRED
-------------------------------------------------------------------------------
Interfaces : 6
===============================================================================
A:COMMERCIAL_AGG# show router 10000 interface

===============================================================================
Interface Table (Service: 10000)
===============================================================================
Interface-Name                   Adm       Opr(v4/v6)  Mode    Port/SapId
   IP-Address                                                   PfxState
-------------------------------------------------------------------------------
TPWODL-ICC-AC-R01                Up        Up/Down     VPRN    lag-5:110
   10.190.199.60/31                                             n/a
loopback-1                       Up        Up/Up       VPRN    loopback
   10.190.204.17/32                                             n/a
   fe80::43:4033:57df:6d0a/64                                   PREFERRED
-------------------------------------------------------------------------------
Interfaces : 2
===============================================================================
A:COMMERCIAL_AGG# show router 10000 interfac
```

VRF FOR IT AND OT

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

www.isuw.in

Links/References (If any)